Olympiad Test Paper — Science (Class 7)

Chapter 1: Nutrition in Plants

Subject	Science (NCERT Class 7)
Chapter	1 — Nutrition in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. The word equation for photosynthesis is: carbon dioxide + water → (in the presence of sunlight and chlorophyll) → glucose + oxygen. Most questions need reasoning or careful application — watch for traps based on common misconceptions. Do not write on this paper — mark answers on your answer sheet.

---

1. An organism that builds its own food from simple inorganic raw materials using an outside energy source is called:

(A) a heterotroph
(B) an autotroph
(C) a saprotroph
(D) a parasite

2. A child argues, "Plants must get their food from the soil, because that is where their roots are." The single best piece of evidence against this is that:

(A) lettuce can be grown to full size in water-only nutrient baths with no soil at all
(B) roots are usually white, not green
(C) soil contains many minerals
(D) plants also need sunlight

3. In the word equation for photosynthesis, the two raw materials taken in are:

(A) glucose and oxygen
(B) oxygen and water
(C) carbon dioxide and water
(D) glucose and carbon dioxide

4. The two products formed during photosynthesis are:

(A) carbon dioxide and water
(B) glucose and oxygen
(C) water and oxygen
(D) glucose and carbon dioxide

5. Leaves look green because chlorophyll:

(A) absorbs green light strongly
(B) is made of green sugar
(C) reflects green light while absorbing red and blue
(D) turns sunlight into a green colour

6. A potted plant is sealed inside a clear glass jar and left in **bright sunlight** for a day. Compared with the open air, the air inside the jar will become:

(A) richer in oxygen than ordinary air (B) pure carbon dioxide

(C) richer in carbon dioxide than ordinary air(D) completely free of all gases

7. The same sealed jar with the plant is now kept in a **completely dark** cupboard for a day. The air inside will become:

(A) richer in carbon dioxide, because the plant only respires

(B) richer in oxygen, because the plant still photosynthesises

(C) unchanged, because plants do nothing in the dark

(D) pure oxygen

8. A leaf is tested with iodine after a few hours in sunlight and turns blue-black. This proves the leaf has made:

(A) protein (B) chlorophyll

(C) oxygen (D) starch

9. Before the starch test, a green leaf is first boiled in alcohol. The purpose of this step is to:

(A) add starch to the leaf (B) make the leaf greener

(C) kill insects on the leaf

(D) remove the green chlorophyll so the colour change is easy to see

10. The tiny pores on a leaf through which carbon dioxide enters and oxygen leaves are the:

(A) stomata (B) xylem

(C) phloem (D) chlorophyll

11. Cuscuta (amarbel) is a pale, yellow-orange climbing plant with no green colour. From this we can correctly conclude that Cuscuta:

(A) photosynthesises faster than green plants(B) makes its food at night

(C) is a kind of fungus

(D) has very little chlorophyll and steals food from a host plant

12. A pitcher plant grows in soil that is very poor in one mineral, so it traps and digests insects. The insect provides the plant mainly with:

(A) energy, because the plant cannot photosynthesise

(B) nitrogen (C) carbon dioxide

(D) water

13. A student says, "Venus flytraps catch insects because they cannot make their own food." The correct reply is:

(A) true — they have no chlorophyll
(B) true — they get all their energy from insects
(C) false — they have green leaves and still photosynthesise; insects only supply nitrogen
(D) false — they actually eat soil

14. Mushrooms growing on a rotting log obtain food by:

(A) photosynthesis in their caps
(B) secreting digestive juices onto the dead wood and absorbing the digested matter
(C) catching insects
(D) taking food from a living host

15. A lichen growing on a bare rock is a partnership between:

(A) two fungi
(B) two algae
(C) a fungus and an alga
(D) a plant and an animal

16. In the lichen partnership, the **food-making** job is done by the:

(A) fungus
(B) alga
(C) rock
(D) neither — it absorbs food from rain

17. Farmers grow pulses (legumes) and then plough them back into a field because the bacteria in their root nodules:

(A) eat harmful insects
(B) make the soil greener
(C) fix nitrogen from the air into the soil
(D) absorb extra water

18. The bacterium that lives in the root nodules of legumes and enriches the soil with nitrogen is:

(A) Rhizobium
(B) Cuscuta
(C) chlorophyll
(D) amoeba

19. Green plants are called the **producers** of a food chain because they:

(A) eat the most food
(B) are the largest organisms
(C) make the food on which all other organisms ultimately depend
(D) produce the most oxygen at night

20. Which of these is the correct order of who depends on whom?

(A) grass → goat → tiger
(B) goat → grass → tiger
(C) tiger → goat → grass
(D) grass → tiger → goat

21. A green leaf, a green stem of a small plant, and a white (non-green) flower petal are each tested for starch after a day in the sun. Which is/are likely to turn blue-black?

(A) only the leaf
(B) none of them
(C) all three
(D) the leaf and the green stem

22. A plant is given everything it needs **except** that one leaf is partly covered with black paper. After a day in the sun, the iodine test on that leaf shows:

(A) the whole leaf turns blue-black (B) only the covered part turns blue-black

(C) only the uncovered (light-exposed) part turns blue-black

(D) no part of the leaf turns blue-black

23. Why do most leaves have a broad, flat, thin shape?

(A) to store large amounts of water

(B) to catch the maximum sunlight over a large surface

(C) to make them heavy so they do not blow away

(D) to keep insects away

24. Water absorbed by the roots is carried up to the leaves through the:

(A) stomata (B) phloem

(C) chlorophyll (D) xylem

25. The food (glucose) made in the leaves is carried to the rest of the plant through the:

(A) xylem (B) phloem

(C) stomata (D) roots

26. Some desert plants have green stems and very few or no leaves. These green stems:

(A) cannot photosynthesise at all (B) take over the job of photosynthesis

(C) only store water and never make food(D) are actually parasites

27. During the **day**, a healthy green plant in sunlight:

(A) only takes in oxygen

(B) does neither photosynthesis nor respiration

(C) photosynthesises and respires, with a large net release of oxygen

(D) only releases carbon dioxide

28. At **night**, the same plant:

(A) photosynthesises faster than in the day(B) stops all life processes

(C) releases a large amount of oxygen

(D) only respires — taking in oxygen and giving out carbon dioxide

29. A pot of soil is sterilised so it has no Rhizobium or other nitrogen-fixing bacteria, and a bean plant is grown in it with plenty of water and sunlight but no fertiliser. Over weeks the plant is most likely to:

(A) grow faster than normal (B) show poor growth from lack of nitrogen

(C) die immediately for lack of carbon dioxide(D) turn into a parasite

30. Which gas do plants take **in** through their stomata for photosynthesis?

(A) oxygen (B) nitrogen

(C) carbon dioxide (D) hydrogen

31. The energy that drives photosynthesis comes from:

(A) sunlight (B) the soil
(C) the minerals in fertiliser (D) the warmth of the plant's own body

32. A teacher covers a leaf with a transparent plastic bag for a day in the sun. Tiny water droplets collect inside the bag. These mainly come from the leaf through:

(A) the xylem tubes opening at the surface (B) water vapour escaping from the stomata
(C) the roots (D) melting chlorophyll

33. Which one of the following is an autotroph?

(A) a mushroom (B) a Cuscuta vine
(C) a mango tree
(D) a Venus flytrap getting its nitrogen from insects

34. A heterotroph is best described as an organism that:

(A) makes its own food from sunlight (B) has chlorophyll in every cell
(C) lives only in water (D) depends on other organisms for its food

35. Cuscuta is classified as a **parasite** rather than a saprotroph because it:

(A) feeds on dead and decaying matter (B) makes its own food
(C) takes food from a **living** host plant, harming it
(D) catches insects

36. Repeatedly growing the same crop drains the soil of one nutrient in particular, which farmers then replace with manure or fertilisers. That nutrient is:

(A) oxygen (B) carbon
(C) glucose (D) nitrogen

37. Although air is about 78% nitrogen gas, most plants cannot use it directly because:

(A) the nitrogen in air is the wrong colour
(B) plants cannot absorb nitrogen gas directly; it must first be fixed into the soil
(C) plants do not need nitrogen at all (D) leaves have no stomata for nitrogen

38. Which statement is **incorrect**?

(A) Autotrophs make their own food.
(B) Saprotrophs feed on dead and decaying matter.
(C) Parasites make their own food and give some to a host.
(D) Insectivorous plants trap insects for nitrogen.

39. A bean seed is sprouted on damp cotton wool in a dark cupboard. For the first several days it still grows a shoot. This is possible because:

(A) it photosynthesises in the dark (B) it lives off food stored inside the seed
(C) the cotton wool feeds it (D) it becomes a saprotroph

40. Which combination of conditions is **necessary** for a green leaf to photosynthesise?

(A) sunlight, chlorophyll, carbon dioxide and water

(B) sunlight, oxygen, glucose and water (C) darkness, chlorophyll and oxygen

(D) chlorophyll, glucose and nitrogen

41. A green plant is kept in a closed box that is supplied with light but from which **all carbon dioxide has been removed** (absorbed by a chemical). The leaf is then tested with iodine. The result is:

(A) it does not turn blue-black, because a raw material (CO_2) was missing

(B) it turns blue-black, because light alone makes starch

(C) it turns blue-black, because chlorophyll is present

(D) it turns red

42. The chief difference in nutrition between a green plant and an animal is that:

(A) the plant eats other organisms; the animal makes its own food

(B) the plant makes its own food; the animal must obtain ready-made food

(C) both make their own food (D) neither needs food

43. Extra glucose made in the leaves is stored in many plants in the form of:

(A) oxygen (B) carbon dioxide

(C) chlorophyll (D) starch

44. A potato tuber and a wheat grain both turn deep blue-black with iodine. This tells us they are rich stores of:

(A) protein (B) fat

(C) starch (D) chlorophyll

45. Two identical green plants are kept in sunlight. Plant X is in normal air; plant Y is in air from which carbon dioxide has been removed. After a day, the starch test will be positive in:

(A) both plants (B) plant Y only

(C) neither plant (D) plant X only

46. Which of the following correctly pairs an organism with its mode of nutrition?

(A) mushroom — autotroph (B) Cuscuta — saprotroph

(C) mango tree — parasite

(D) Rhizobium in legume roots — symbiotic (mutually helpful) partner

47. In a sealed terrarium kept in good light, a small plant can survive for a long time because the gases roughly balance: the plant's photosynthesis uses the ___ that its own respiration releases, and respiration uses the oxygen that photosynthesis releases. The blank is:

(A) oxygen (B) nitrogen

(C) carbon dioxide (D) glucose

48. Which is the most accurate statement about a leaf's green colour and its food-making?

(A) A leaf must be green to carry out photosynthesis, because chlorophyll does the light-capturing.

(B) Any colour of leaf photosynthesises equally well.

(C) Green leaves cannot photosynthesise.

(D) Colour has nothing to do with photosynthesis.

49. A red-leaved copper beech tree still grows tall and healthy in full sun. The best explanation is that:

(A) it has chlorophyll, but a red pigment masks the green colour

(B) it does not need to photosynthesise (C) it gets all its food from the soil

(D) it is a parasite

50. Place these in the correct sequence of a green plant making and using food:

(A) glucose made → carbon dioxide and water taken in → starch stored

(B) carbon dioxide and water taken in → glucose made in light → starch stored

(C) starch stored → glucose made → raw materials taken in

(D) oxygen taken in → glucose made → starch stored

51. A gardener notices a Cuscuta vine has wrapped around her hibiscus and the hibiscus is wilting. The Cuscuta is acting as a:

(A) saprotroph helping the hibiscus (B) symbiotic partner

(C) producer (D) parasite harming its host

52. Which experiment best shows that **sunlight** is needed for photosynthesis?

(A) cover part of a leaf with black paper, then do the starch test on the whole leaf

(B) keep a plant in coloured water (C) count the stomata on a leaf

(D) weigh a leaf before and after a day

53. Which experiment best shows that **chlorophyll** is needed for photosynthesis?

(A) keep a plant in the dark for two days (B) put a leaf in iodine straightaway

(C) seal a plant in a jar

(D) do the starch test on a variegated (part-green, part-white) leaf and compare the parts

54. On a variegated leaf (green at the edges, white in the middle) left in sunlight, the iodine starch test turns blue-black:

(A) only in the white (non-green) middle (B) only in the green parts

(C) over the whole leaf equally (D) nowhere on the leaf

55. The overall reason green plants are vital to nearly all life on Earth is that they:

(A) trap sunlight's energy as food and release the oxygen that other organisms breathe

(B) are very beautiful (C) take up space in forests

(D) only release carbon dioxide

56. A farmer who has just harvested a heavy crop wants to restore soil nitrogen **without** chemical fertiliser. The best choice is to:

(A) grow a legume crop whose roots carry Rhizobium

(B) leave the field bare for a year (C) burn the field

(D) water the field with extra water

57. A houseplant kept in a dim corner away from any window slowly yellows and weakens, even though it is watered. The most likely reason is that:

(A) it is releasing too much oxygen (B) it is drowning in water

(C) it has become a parasite

(D) without enough light it cannot photosynthesise enough food to stay healthy

58. Which statement about insectivorous plants is **true**?

(A) They have no chlorophyll.

(B) They get all their energy from digesting insects.

(C) They grow in nitrogen-poor soil and trap insects to obtain nitrogen, while still photosynthesising.

(D) They are the same as parasitic plants.

59. A sealed jar contains a green plant **and** a small lit candle, kept in bright light. The plant can help keep the candle burning longer than an empty sealed jar would, because the plant:

(A) supplies oxygen, which a flame needs, through photosynthesis

(B) supplies carbon dioxide that feeds the flame

(C) puts the flame out faster (D) absorbs the candle's heat

60. Which single statement best captures the central idea of the chapter "Nutrition in Plants"?

(A) All plants get their food ready-made from the soil.

(B) Only fungi can make food from sunlight.

(C) Plants and animals make food in exactly the same way.

(D) Green plants make their own food by photosynthesis, while some plants and other organisms feed in special ways (parasitic, insectivorous, saprotrophic, symbiotic).

End of question paper. Maximum marks: 240. Marking: +4 correct, –1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 1: Nutrition in Plants

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	D	31	A	41	A	51	D
2	A	12	B	22	C	32	B	42	B	52	A
3	C	13	C	23	B	33	C	43	D	53	D
4	B	14	B	24	D	34	D	44	C	54	B
5	C	15	C	25	B	35	C	45	D	55	A
6	A	16	B	26	B	36	D	46	D	56	A
7	A	17	C	27	C	37	B	47	C	57	D
8	D	18	A	28	D	38	C	48	A	58	C
9	D	19	C	29	B	39	B	49	A	59	A
10	A	20	A	30	C	40	A	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) An organism that builds its own food from simple inorganic raw materials (CO_2, water, minerals) using an outside energy source (sunlight) is an *autotroph* (auto = self). A heterotroph (A) eats others; a saprotroph (C) feeds on dead matter; a parasite (D) takes food from a living host — all three depend on ready-made food.

2. (A) "Food from soil" is the classic misconception. The decisive evidence is hydroponics: lettuce (and many crops) can be grown to full size in water-only nutrient baths with *no soil at all*. If food came from soil, this would be impossible. White roots (B), the presence of minerals (C), and needing sunlight (D) are all true facts but do not directly refute the "food comes from soil" claim — only the no-soil-grown plant does.

3. (C) Photosynthesis takes *in* carbon dioxide and water as raw materials. Glucose and oxygen (A) are the products, not the inputs; (B) and (D) each mix a product with a raw material.

4. (B) The products of photosynthesis are glucose (food) and oxygen. Carbon dioxide and water (A) are the raw materials; (C) and (D) again confuse inputs with outputs.

5. (C) Chlorophyll *reflects* green light (which is why we see green) while *absorbing* the red and blue light it uses to power photosynthesis. It does not absorb green strongly (A) — if it did, leaves would not look green; (B) and (D) are nonsense distractors.

6. (A) In bright sunlight photosynthesis runs strongly, consuming CO_2 and releasing O_2 at a rate that far exceeds the plant's own respiration. So the sealed air becomes *richer in oxygen* than ordinary air. (C) reverses the gases (that is the dark case); (B) and (D) are impossible.

7. (A) In complete darkness there is no photosynthesis, but the plant still *respires* — taking in O_2 and giving out CO_2 — so the sealed air becomes richer in carbon dioxide. (B) wrongly assumes photosynthesis in the dark; (C) wrongly assumes plants are inert at night; (D) is impossible.

8. (D) Iodine turns blue-black in the presence of *starch*. A blue-black leaf therefore shows the leaf has made starch (from the glucose of photosynthesis). Iodine does not test for protein (A), chlorophyll (B), or oxygen (C).

9. (D) Boiling the leaf in alcohol dissolves out the green chlorophyll, decolourising the leaf so the blue-black starch colour is easy to see against a pale background. It does not add starch (A), make the leaf greener (B), or kill insects (C) — in fact it removes the green.

10. (A) The tiny pores that let CO_2 in and O_2 out are the *stomata* (mostly on the leaf underside). Xylem (B) carries water, phloem (C) carries food, and chlorophyll (D) is a pigment, not a pore.

11. (D) Green colour means chlorophyll; a pale, yellow-orange plant has little or none and therefore cannot make its own food, so it must steal food from a host — i.e. it is a parasite. It cannot photosynthesise faster (A) or "at night" (B) without chlorophyll, and lacking chlorophyll does not make it a fungus (C); Cuscuta is still a plant.

12. (B) Pitcher plants grow in nitrogen-poor soil, so they trap insects to obtain *nitrogen*, not energy. Trap (A): they are green and still photosynthesise for their energy. They get CO_2 from air (C) and water from soil (D) like any plant.

13. (C) The student's claim is false. Venus flytraps have green leaves and photosynthesise for their energy; they catch insects only to obtain *nitrogen* missing from their boggy soil. (A) and (B) repeat the "they can't make food / get energy from insects" misconception; (D) is wrong — they do not eat soil.

14. (B) Mushrooms are fungi (saprotrophs): they grow thread-like hyphae into dead matter, secrete digestive juices *outside* the body, and absorb the digested food. They have no chlorophyll, so (A) is wrong; they do not trap insects (C) or feed on a living host (D) — the wood here is dead.

15. (C) A lichen is a symbiotic partnership of a *fungus* (which provides shelter and absorbs water/minerals) and an *alga* (which makes food by photosynthesis). It is not two fungi (A), two algae (B), or a plant-and-animal pair (D).

16. (B) Within a lichen, the green *alga* contains chlorophyll and does the food-making (photosynthesis); the fungus provides housing and water. So the alga, not the fungus (A) or the rock (C), makes the food; (D) is wrong because food is made, not absorbed from rain.

17. (C) Legume root nodules house *Rhizobium*, which fixes nitrogen from the air into a form plants can use, enriching the soil. Ploughing the legumes back returns this nitrogen to the field. The bacteria do not eat insects (A), simply "green" the soil (B), or absorb water (D).

18. (A) The nitrogen-fixing bacterium of legume root nodules is *Rhizobium*. Cuscuta (B) is a parasitic plant, chlorophyll (C) is a pigment, and amoeba (D) is a protozoan — none fix nitrogen in legume roots.

19. (C) Green plants are called *producers* because they make the food (using sunlight) on which every other organism in a food chain ultimately depends. They do not necessarily eat the most (A) or grow the largest (B); and they release oxygen by day, not "at night" (D).

20. (A) A food chain runs producer → primary consumer → higher consumer: grass (producer) → goat (eats grass) → tiger (eats goat). The other orders put a consumer before its food, which is impossible.

21. (D) After a day in the sun, the starch test is positive wherever chlorophyll allowed photosynthesis. Both the green leaf and the green stem contain chlorophyll, so both make starch and turn blue-black; the white (non-green) petal has no chlorophyll, so it cannot make starch and stays pale. Hence *the leaf and the green stem* test positive. (A) forgets the green stem also photosynthesises; (C) wrongly includes the white petal; (B) "none" ignores that green parts do make starch.

22. (C) Photosynthesis (and hence starch) needs light. The covered part got no light, so only the *uncovered, light-exposed* part makes starch and turns blue-black. (A) ignores the covering; (B) reverses it; (D) wrongly says no starch forms at all.

23. (B) A broad, flat, thin leaf exposes a large surface to the sun, catching the maximum sunlight for photosynthesis (and the thinness lets light and gases reach all cells). It is not for storing water (A), for weight (C), or to repel insects (D).

24. (D) Water absorbed by the roots travels up to the leaves through the *xylem*. Stomata (A) are gas pores, phloem (B) carries food downward/outward, and chlorophyll (C) is a pigment.

25. (B) The glucose made in the leaves is carried to the rest of the plant through the *phloem*. Xylem (A) carries water *up*; stomata (C) are gas pores; roots (D) absorb water, they do not transport made food.

26. (B) When a plant has green stems and few leaves (as in many desert plants), the green stems contain chlorophyll and *take over photosynthesis*. So (A) is wrong; they do more than store water (C); and having green stems does not make them parasites (D).

27. (C) By day a green plant *both* photosynthesises and respires. Photosynthesis far outpaces respiration, so there is a large *net release of oxygen*. (A) and (D) describe only respiration; (B) wrongly says it does nothing.

28. (D) At night there is no light, so photosynthesis stops and the plant *only respires* — taking in O_2 and giving out CO_2, just like animals. It does not photosynthesise faster (A), release lots of O_2 (C), or stop all life processes (B) — respiration continues.

29. (B) With no nitrogen-fixing bacteria and no fertiliser, the soil cannot supply nitrogen, which plants need to build proteins. The plant will show *poor growth from lack of nitrogen*. It will not grow faster (A); it gets CO_2 from the air, so it will not "die for lack of CO_2" (C); and plants do not turn into parasites (D).

30. (C) Plants take *in* carbon dioxide through their stomata for photosynthesis. They release O_2 as a product (A); atmospheric nitrogen gas (B) cannot be used directly; hydrogen (D) is not involved.

31. (A) The energy that powers photosynthesis comes from *sunlight*, captured by chlorophyll. It does not come from the soil (B), from fertiliser minerals (C), or from the plant's own warmth (D).

32. (B) Droplets inside the bag come from water vapour the leaf releases (transpiration) through its *stomata*, which then condenses on the cool plastic. Xylem (A) does not open at the surface; the roots (C) are not bagged; chlorophyll does not "melt" (D).

33. (C) An autotroph makes its own food by photosynthesis: the *mango tree* qualifies. A mushroom (A) is a saprotroph, a Cuscuta vine (B) is a parasite, and a Venus flytrap getting nitrogen from insects (D) — though green — is described here in its insect-trapping (heterotrophic supplement) role, making the mango tree the clean autotroph answer.

34. (D) A heterotroph cannot make its own food and so *depends on other organisms* for it. Making food from sunlight (A) describes an autotroph; living in water (B) is irrelevant; having chlorophyll in every cell (C) is false even for plants and would point to an autotroph.

35. (C) Cuscuta is a *parasite* because it takes food from a *living* host plant and harms it. A saprotroph feeds on *dead* matter (A); Cuscuta does not make its own food (B) — it lacks chlorophyll; and it does not catch insects (D).

36. (D) Repeatedly cropping the same field drains *nitrogen* most of all, which farmers replace with manure or fertilisers (or by growing legumes). Oxygen (A) and carbon (B) come from air; glucose (C) is food the plant makes itself, not a soil nutrient.

37. (B) Although air is ~78% nitrogen gas, plants *cannot absorb nitrogen gas directly*; it must first be "fixed" into soil compounds (by Rhizobium or fertilisers) before roots can take it up. The colour claim (A) is silly; plants do need nitrogen (C); and stomata handle CO_2/O_2, not nitrogen uptake (D).

38. (C) The incorrect statement is "Parasites make their own food and give some to a host." Parasites do the opposite — they *take* food from a host and harm it. The other three statements (autotrophs make their own food; saprotrophs feed on dead matter; insectivorous plants trap insects for nitrogen) are all correct.

39. (B) A seed sprouting in the dark grows by living off the *food stored inside the seed* (the cotyledons). It cannot photosynthesise in the dark (A); the cotton wool only holds moisture, it is not food (C); and it does not become a saprotroph (D).

40. (A) Photosynthesis needs four things together: *sunlight, chlorophyll, carbon dioxide and water*. Option B lists oxygen and glucose, which are *products*; C demands darkness (wrong); D includes glucose (a product) and nitrogen (a mineral, not a photosynthesis input).

41. (A) Carbon dioxide is a raw material. With all CO_2 removed, no glucose (and hence no starch) can form even in light, so the iodine test does *not* turn blue-black. (B) wrongly says light alone suffices; (C) ignores the missing raw material; (D) is not what iodine does with starch absent.

42. (B) The key nutritional difference is that the *plant makes its own food* (autotroph) while the *animal must obtain ready-made food* (heterotroph). Option A reverses this; C is false (animals cannot make food); D is false (both need food).

43. (D) Extra glucose made in the leaves is stored as *starch* (insoluble, easy to stockpile). Oxygen (A) and carbon dioxide (B) are gases involved in the reaction, not storage forms; chlorophyll (C) is a pigment, not stored food.

44. (C) Iodine turns blue-black only with *starch*, so a deep blue-black potato and wheat grain are rich starch stores. The test does not detect protein (A), fat (B), or chlorophyll (D).

45. (D) Plant X (normal air) has CO_2, makes starch → positive. Plant Y (CO_2 removed) lacks a raw material, makes no starch → negative. So the starch test is positive in *plant X only*. (A) and (B) wrongly include Y; (C) wrongly excludes X.

46. (D) The correct pairing is *Rhizobium in legume roots — symbiotic partner* (the bacterium and plant both benefit). A mushroom is a saprotroph, not an autotroph (A); a mango tree is an autotroph, not a parasite (C); and Cuscuta is a parasite, not a saprotroph (B).

47. (C) In a sealed, lit terrarium the gases balance: the plant's *photosynthesis* uses the *carbon dioxide* released by its own *respiration*, while respiration uses the oxygen photosynthesis makes. So the blank is carbon dioxide. Oxygen (A) is the gas going the other way; nitrogen (B) and glucose (D) are not part of this gas exchange loop.

48. (A) Because chlorophyll is what captures light, a leaf must contain chlorophyll (be functionally green) to photosynthesise — so "a leaf must be green ... because chlorophyll does the light-capturing" is the accurate statement. (B) is false (a white, chlorophyll-free part cannot); (C) is false; (D) ignores chlorophyll's central role.

49. (A) A copper beech is healthy because it still *has chlorophyll*; a red pigment (anthocyanin) simply *masks* the green. It does photosynthesise (so A is right, B wrong); it does not get food from soil (C); and it is not a parasite (D).

50. (B) The correct order is: *raw materials (CO_2 + water) taken in → glucose made in light → starch stored*. Option A puts glucose before its raw materials; C reverses the whole sequence; D starts with oxygen being taken in, which is respiration, not photosynthesis.

51. (D) A Cuscuta vine taking food from a living, wilting hibiscus is a *parasite harming its host*. It is not a saprotroph helping anything (A), not a mutually beneficial symbiotic partner (B), and not a producer (C) — it makes no food of its own.

52. (A) To show *sunlight* is needed, you keep everything else equal and vary only light: cover part of a leaf with black paper, then do the starch test — only the lit part turns blue-black. Coloured water (B) shows xylem transport; counting stomata (C) and weighing (D) test other things.

53. (D) To show *chlorophyll* is needed, use a *variegated leaf* (part green, part white) where light and water are identical for both parts — only the green (chlorophyll-containing) part makes starch. Darkness (A) tests light, not chlorophyll; iodine straightaway (B) tests nothing controlled; sealing a jar (C) tests gases.

54. (B) On a variegated leaf, only the green (chlorophyll-bearing) parts can photosynthesise, so the iodine test turns blue-black *only in the green parts*. The white middle (A) has no chlorophyll and stays pale; (C) and (D) are therefore wrong.

55. (A) Green plants are vital because they *trap sunlight's energy as food and release the oxygen* that other organisms breathe — the dual basis of almost all life. Being beautiful (B) or taking up space (C) is trivial, and they release oxygen (not only CO_2), so (D) is false.

56. (A) To restore soil nitrogen without chemical fertiliser, *grow a legume crop whose roots carry Rhizobium*, which fixes atmospheric nitrogen into the soil. Leaving the field bare (B) adds no nitrogen; burning (C) destroys organic matter; extra water (D) does not supply nitrogen.

57. (D) In a dim corner the plant cannot get *enough light to photosynthesise* enough food, so it weakens and yellows despite watering. Releasing oxygen does not harm it (A); it is not drowning (B is a different problem and not implied); and it does not become a parasite (C).

58. (C) Insectivorous plants *grow in nitrogen-poor soil and trap insects to obtain nitrogen, while still photosynthesising* for energy. They do have chlorophyll (A is false), do not get all energy from insects (B is false), and are not the same as parasites, which take food from a living host (D is false).

59. (A) In bright light the plant photosynthesises and releases *oxygen*, which a flame needs to keep burning — so the plant helps the candle last longer than in an empty sealed jar. A flame is not "fed" by CO_2 (B); the plant does not put the flame out faster (C) or matter here by absorbing heat (D).

60. (D) The chapter's central idea: *green plants make their own food by photosynthesis, while some plants and other organisms feed in special ways (parasitic, insectivorous, saprotrophic, symbiotic)*. (A) repeats the "food from soil" misconception; (B) is false; (C) wrongly equates plant and animal nutrition.

Solutions complete: 60 / 60.